First-Passage Linear Transport for Almost-All Collatz Descent

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Abstract

Let

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ (3n+1)/2, & n \equiv 1 \pmod 2, \end{cases} \qquad T_{\min}(n) = \min_{k \ge 0} T^k(n).$$

For every fixed $0 < \delta < 1$, we prove

$$T_{\min}(n) \le \exp\left((\log n)^{1-\delta} \right)$$

on a set of natural density one. More quantitatively, for every fixed $0 < \sigma < 1 - \delta$, the exceptional count up to X is at most

$$5X \exp(-c_{\delta,\sigma}(\log X)^{\sigma})$$

for all sufficiently large X. The descent is witnessed before $6.953 \log n$ shortcut steps and before $10.44 \log n$ raw Collatz steps, and the shortcut orbit up to that witness may be confined below $n^{1+\beta}$ for any separately fixed $\beta > 0$. For every fixed $0 < \alpha < 1$ and $\varepsilon > 0$, a sharper graded clock reaches n^{α} before

$$\left(\frac{2(1-\alpha)}{\log(4/3)} + \varepsilon \right) \log n$$

shortcut steps. The fixed-power exceptional set also satisfies $5X \exp(-c_{\alpha,\sigma}(\log X)^{\sigma})$ for every fixed $0 < \sigma < 1$.

The proof is elementary once the parity-vector bijection is available. A maximal parity barrier supplies a dense set on which every logarithmic shell enters a lower power scale. Exact reversal at the first entrance forces all sources in a tagged landing fiber to have one common odd count. This gives linear transport for every target subset of the landing shell. A totalized stopped map then iterates the transport through $O(\log \log n)$ scales.

1. Introduction and main results

We work throughout with the shortcut Collatz map T displayed in the abstract. All unqualified logarithms are natural, and $\log_2$ denotes the base-two logarithm. Here $\mathbb{N} = \{1, 2, 3, \dots\}$. Put

$$a_0 = \frac{\log_2 3}{2}, \qquad \varrho = \frac{\sqrt{3}}{2}. \tag{1.1}$$

The proof below is organized around the first entrance below a shell-scaled target. It does not estimate a fixed-time endpoint distribution: the target set in the transport theorem is arbitrary, and the stopped map is closed by an identity branch on a finite startup interval.

This is a standalone V2 argument. No fixed-time endpoint transport theorem is a proof dependency of any result below. A companion formalization accompanies this version and is not used as manuscript evidence.

Theorem 1.1 (stretched-logarithmic natural-density descent)

For every fixed $0 < \delta < 1$,

$$T_{\min}(n) \le \exp((\log n)^{1-\delta}) \tag{1.2}$$

on a set of natural density one. The endpoint $\delta = 1$ is not asserted.

Corollary 1.2 (quantitative exceptional count)

For every fixed $0 < \delta < 1$ and every $0 < \sigma < 1 - \delta$, there are constants $c_{\delta,\sigma} > 0$ and $X_{\delta,\sigma}$ such that

$$\#\{1 \le n \le X : T_{\min}(n) > \exp((\log n)^{1-\delta})\} \le 5X \exp(-c_{\delta,\sigma}(\log X)^{\sigma}) \tag{1.3}$$

for every $X \ge X_{\delta,\sigma}$.

Corollary 1.3 (time and orbit ceiling)

For every fixed $0 < \delta < 1$, the density-one set in Theorem 1.1 may be chosen so that every sufficiently large retained n has an integer

$$0 \le k < 6.953 \log n \tag{1.4}$$

with

$$T^k(n) \le \exp((\log n)^{1-\delta}). \tag{1.5}$$

For every separately fixed $\beta > 0$, the set and witness may also be chosen so that

$$\max_{0 \le j \le k} T^j(n) \le n^{1+\beta}. \tag{1.6}$$

To compare clocks with papers using the raw Collatz map, put

$$\mathrm{Col}(n) = \begin{cases} n/2, & n \equiv 0 \pmod 2, \\ 3n+1, & n \equiv 1 \pmod 2. \end{cases}$$

For an odd input, one shortcut step is two raw steps; for an even input it is one. The shortcut orbit deletes only intermediate values $3m + 1 > m$, so $T_{\min}(n) = \mathrm{Col}_{\min}(n)$. The maximal parity barrier retains enough pointwise odd-count information to sharpen the worst-case factor two: the witness in Corollary 1.3 also occurs before $10.44 \log n$ raw steps.

Corollary 1.4 (timed fixed-power descent)

For every fixed $\alpha > 0$, there is a set S_α of natural density one such that every sufficiently large $n \in S_\alpha$ has an integer k satisfying

$$0 \le k < 6.953 \log n, \qquad T^k(n) \le n^\alpha.$$

Corollary 1.5 (quantitative fixed-power exceptional count)

For every fixed $\alpha > 0$ and every fixed $0 < \sigma < 1$, there are constants $c_{\alpha,\sigma} > 0$ and $X_{\alpha,\sigma}$ such that

$$\#\{1 \le n \le X : T_{\min}(n) > n^\alpha\} \le 5X \exp(-c_{\alpha,\sigma}(\log X)^\sigma) \tag{1.7}$$

for every $X \ge X_{\alpha,\sigma}$.

Corollary 1.6 (smooth graded shortcut clock)

For every fixed $0 < \alpha < 1$ and every fixed $\varepsilon > 0$, there is a set $S_{\alpha,\varepsilon}$ of natural density one such that every sufficiently large $n \in S_{\alpha,\varepsilon}$ has an integer k with

$$0 \le k < \left(\frac{2(1-\alpha)}{\log(4/3)} + \varepsilon\right) \log n, \qquad T^k(n) \le n^\alpha. \tag{1.8}$$

Relation to previous almost-all results

The theorem should be read along several independent axes: density notion, target scale, clock, and exceptional rate. The closest comparisons are:

Result	Status	Density	Target reached for almost all starts	Clock retained in the stated result	Quantitative exception
Korec [3]	published	natural	n^θ, every fixed $\theta > a_0$	not part of the cited theorem	not used here
Inselmann [2]	preprint	natural	n^ε, every fixed $\varepsilon > 0$	$2\log n/\log(4/3)$ shortcut steps	density convergence
Tao [5]	published	logarithmic	every $f(n) \to \infty$	no single global clock in the headline theorem	logarithmic-density estimate
Mazur [4]	manuscript with a pinned formal artifact	natural	every $f(n) \to \infty$	$< 436 \log n$ raw steps	fixed-target logarithmic rate
Allikvere [1]	preprint	natural	every $f(n) \to \infty$	$< 12 \log n$ raw steps	$O((\log N_0)^{-1/29} + X^{-1/2000})$ for a fixed target
This paper	standalone V2	natural	$\exp((\log n)^{1-\delta})$, every fixed $0 < \delta < 1$	$< 6.953 \log n$ shortcut steps and $< 10.44 \log n$ raw steps; graded shortcut clock (1.8) for fixed powers	$5X\exp(-c(\log X)^\sigma)$, every fixed $0 < \sigma < 1-\delta$; every fixed $0 < \sigma < 1$ for fixed powers

Thus the present target is smaller than every fixed power target, and hence gives a stronger descent conclusion on that axis, but it is weaker than an arbitrary diverging function. In particular, this paper does not claim to supersede the arbitrary-threshold conclusions of [1] or [4], and its shortcut clock is on the scale already identified in [2]. Its distinct contribution is a short, self-contained route to stretched-logarithmic natural-density descent, a stretched-exponential-in-log exceptional rate for that target, and a pointwise first-passage transport estimate valid for every landing subset. The last statement removes any need to prove that a target generated by earlier blocks resembles a fresh random set.

The proof has four load-bearing steps.

1. The parity-vector bijection turns a complete dyadic shell into the uniform Boolean cube.
2. A maximal Boolean-walk estimate gives a power-dense all-prefix set.

3. First-passage reversal gives a pointwise tagged-fiber bound and hence linear transport for arbitrary targets.
4. A totalized stopped-map pullback iterates this transport without assuming that a generated target resembles a fresh random set.

2. Density, parity, and affine iterates

For $S \subseteq N$, put

$$B_S(X) = \#\{1 \le n \le X : n \notin S\}.$$

We call S (C, D)**-dense** if

$$B_S(X) \le CX^{1-D} \qquad (X \ge 1), \tag{2.1}$$

where $C > 0$ and $0 < D \le 1$. Every such set has natural density one.

We use the following varying-rate shell summation.

Lemma 2.1 (varying-rate dyadic summation)

Let $0 < \sigma \le 1$, $A, \gamma > 0$, and suppose that, for all sufficiently large M,

$$\#(E \cap [2^M, 2^{M+1})) \le Ae^{-\gamma(M+4)^\sigma} 2^M. \tag{2.2}$$

Then there are $\gamma' > 0$ and X_0 such that

$$\#(E \cap [1, X]) \le (2A + 1)Xe^{-\gamma'(\log X)^\sigma} \qquad (X \ge X_0). \tag{2.3}$$

Proof

Let $2^J \le X < 2^{J+1}$. The shells below $J/2$ contain fewer than $X^{1/2}$ integers. On the remaining shells,

$$M + 4 \ge \frac{\log X}{4 \log 2}.$$

Thus their total contribution is at most

$$2AX \exp\left(-\frac{\gamma}{(4\log 2)^\sigma}(\log X)^\sigma\right).$$

Choose

$$\gamma' = \min\left\{\frac{\gamma}{(4\log 2)^\sigma}, \frac{1}{2}\right\}.$$

For all sufficiently large X, one has $X^{1/2} \le Xe^{-\gamma'(\log X)^\sigma}$. Adding the early and late contributions proves (2.3).
□

For $n \ge 1$, define

$$p_i(n) = 1_{\{T^i(n)\ \mathrm{odd}\}}, \qquad s_k(n) = \sum_{i=0}^{k-1} p_i(n). \tag{2.4}$$

The exact parity coding below is classical in the stopping-time literature; see, for example, Terras [6]. Its short proof is included because no external result is needed by the argument.

Proposition 2.2 (parity-vector bijection)

For every $M \geq 0$,

$$n \bmod 2^M \longmapsto (p_0(n), \dots, p_{M-1}(n)) \tag{2.5}$$

is a bijection from $\mathbb{Z}/2^M\mathbb{Z}$ to $\{0,1\}^M$.

Proof

The relation

$$n \equiv n' \pmod{2^{k+1}} \quad \Longrightarrow \quad T(n) \equiv T(n') \pmod{2^k}$$

shows inductively that the first M parity bits depend only on $n \bmod 2^M$. For the converse, induct on the word length. Suppose the first bit is e, and use the induction hypothesis to recover the residue $m = T(n) \bmod 2^k$ from the remaining k bits. If $e = 0$, the unique compatible residue is

$$n \equiv 2m \pmod{2^{k+1}}.$$

If $e = 1$, it is

$$n \equiv (2m - 1)3^{-1} \pmod{2^{k+1}},$$

where 3 is a unit modulo 2^{k+1}. The first residue is even, the second is odd, and in either case applying T recovers $m \bmod 2^k$. Thus every word has exactly one preimage residue. □

Proposition 2.3 (exact affine iterate)

For every $n \geq 1$ and $k \geq 0$,

$$T^k(n) = \frac{3^{s_k(n)}}{2^k} n + \sum_{i=0}^{k-1} \frac{p_i(n) 3^{s_k(n) - s_{i+1}(n)}}{2^{k-i}}. \tag{2.6}$$

Proof

The assertion is trivial at $k = 0$. Passing from k to $k + 1$, an even letter divides every existing term by two. An odd letter multiplies every existing term by $3/2$ and adds $1/2$. This is exactly the displayed update. □

3. A dense maximal-barrier set

For a parity word of length M, put

$$Y_k = 2s_k - k, \qquad H_M = \max_{0 \leq k \leq M} \left| s_k - \frac{k}{2} \right| = \frac{1}{2} \max_{0 \leq k \leq M} |Y_k|. \tag{3.1}$$

Lemma 3.1 (maximal Boolean-walk bound)

For $M \geq 1$ and $h \geq 0$,

$$2^{-M} \# \{ w \in \{0,1\}^M : H_M(w) > h \} \leq 2 \exp\left(-\frac{2h^2}{M} \right). \tag{3.2}$$

Proof

Fix $\theta > 0$ and stop the Boolean tree at the first prefix u with $|Y(u)| > 2h$. For a frontier node of depth k, set

$$\Phi_\theta(u) = 2^{-k}\cosh(\theta Y(u))(\cosh\theta)^{M-k}.$$

The identity

$$\cosh(\theta(y-1)) + \cosh(\theta(y+1)) = 2\cosh(\theta y)\cosh\theta$$

shows that replacing any unexpanded node by its two children preserves the total potential. Take the frontier to consist of the first-crossing prefixes together with all depth-M words that never cross. Starting from the root, its frontier sum is therefore $(\cosh\theta)^M$. Each crossing node contributes at least $2^{-|u|}\cosh(2\theta h)$, so

$$\Pr(H_M > h) \le \frac{(\cosh\theta)^M}{\cosh(2\theta h)} \le 2\exp\left(\frac{M\theta^2}{2} - 2\theta h\right).$$

Here $\cosh x \le e^{x^2/2}$ and $\cosh x \ge e^{|x|}/2$. Taking $\theta = 2h/M$ proves (3.2); the case $h = 0$ is immediate. □

Define

$$r_k(n) = \sum_{i=0}^{k-1} \frac{p_i(n)}{3^{s_{i+1}(n)}2^{k-i}}, \qquad d_k = s_k - \frac{k}{2}. \tag{3.3}$$

By Proposition 2.3,

$$T^k(n) = \varrho^k n 3^{d_k} + (r_k(n)3^{k/2})3^{d_k}. \tag{3.4}$$

Lemma 3.2 (uniform affine correction)

If $H_M \le h$, then for every $1 \le k \le M$,

$$r_k(n)3^{k/2} \le (2+\sqrt{3})(1-\varrho^k)3^h < (2+\sqrt{3})3^h. \tag{3.5}$$

Proof

For $0 \le i < k$, the barrier gives $s_{i+1} \ge (i+1)/2 - h$. Hence

$$r_k 3^{k/2} \le 3^h \sum_{i=0}^{k-1} \frac{3^{(k-i-1)/2}}{2^{k-i}} = \frac{3^h}{2}\sum_{j=0}^{k-1}\varrho^j,$$

which is (3.5). □

For $0 < \eta \le 1$, let W_η be the set of integers n such that

$$\varrho^k n^{1-\eta} \le T^k(n) \le \varrho^k n^{1+\eta} \qquad (0 \le k \le \lfloor \log_2 n \rfloor). \tag{3.6}$$

Proposition 3.3 (density of the maximal-barrier set)

There are absolute constants $K_W, c_W > 0$ such that W_η is $(K_W, c_W\eta^2)$-dense for every $0 < \eta \le 1$.

Proof

Put $L_3 = \log_2 3$ and, on the shell $I_M = [2^M, 2^{M+1}) \cap \mathbb{N}$, choose

$$h = \frac{\eta M}{2L_3}.$$

Proposition 2.2 and Lemma 3.1 give

$$\frac{\#\{n \in I_M : H_M(n) > h\}}{2^M} \le 2\exp\left(-\frac{\eta^2 M}{2L_3^2}\right). \tag{3.7}$$

Assume $H_M \le h$, $M \ge 4$, and $\eta M \ge 2$. Then

$$3^h = 2^{\eta M/2}, \qquad n^{\eta} \ge 2^{\eta M}, \qquad 2^{\eta M/2} \le \frac{1}{2}n^{\eta}. \tag{3.8}$$

Since $|d_k| \le h$, the multiplicative term in (3.4) lies between $\varrho^k n^{1-\eta}$ and $\frac{1}{2}\varrho^k n^{1+\eta}$: indeed, (3.8) and $n \ge 2^M$ give

$$3^{-h} = 2^{-\eta M/2} \ge n^{-\eta}, \qquad 3^h = 2^{\eta M/2} \le \frac{1}{2}n^{\eta}.$$

By Lemma 3.2, the additive term is at most

$$(2+\sqrt{3})3^{2h} = (2+\sqrt{3})2^{\eta M}.$$

For $k \le M$ and $n \ge 2^M$,

$$\frac{1}{2}\varrho^k n^{1+\eta} \ge \frac{1}{2}\varrho^M 2^{(1+\eta)M} = \frac{1}{2}2^{(a_0+\eta)M}.$$

Because $2^{4a_0} = 9 > 2(2+\sqrt{3})$, this dominates the additive term when $M \ge 4$. Thus (3.6) holds through time M.

If $M < 4$ or $\eta M < 2$, use the trivial shell bound. Consequently there are absolute $K_{sh}, c_0 > 0$, with $c_0 < \log 2$, such that

$$\frac{\#(W_\eta^c \cap I_M)}{2^M} \le K_{sh} e^{-c_0\eta^2 M} \tag{3.9}$$

for every M. Summing the geometric shell series proves (2.1) with

$$D = c_W\eta^2, \qquad c_W = \frac{c_0}{\log 2},$$

and a constant K_W uniform in $0 < \eta \le 1$, because $2e^{-c_0\eta^2} - 1 \ge 2e^{-c_0} - 1 > 0$. □

4. First-passage linear transport

For real $Y > 1$, define

$$\tau_Y(n) = \min\{h \ge 0 : T^h(n) \le Y\} \tag{4.1}$$

when the set is nonempty.

Lemma 4.1 (first-passage band and reverse product)

Let $1 < Y < 2^M$, let $n \in I_M$, and suppose $\tau_Y(n) = h \ge 1$. Write $x_j = T^j(n)$, $y = x_h$, and let s be the number of odd values among $x_0, \ldots, x_{h-1}$. Then

$$\frac{Y}{2} < y \le Y \tag{4.2}$$

and

$$n = \frac{2^h y}{3^s} \prod_{\substack{0 \le j < h \\ x_j \text{ odd}}} \left(1 - \frac{1}{2x_{j+1}}\right). \tag{4.3}$$

Consequently, whenever $h < 2Y$,

$$\left(1 - \frac{h}{2Y}\right) \frac{2^h y}{3^s} \le n \le \frac{2^h y}{3^s}. \tag{4.4}$$

Proof

The final crossing cannot be odd: otherwise $x_h = (3x_{h-1} + 1)/2 > x_{h-1} > Y$. Hence $x_{h-1} = 2y$, proving (4.2). Reversing an even step gives $x_j = 2x_{j+1}$, while reversing an odd step gives

$$x_j = \frac{2x_{j+1} - 1}{3} = \frac{2}{3} x_{j+1} \left(1 - \frac{1}{2x_{j+1}}\right).$$

Multiplication proves (4.3). Every odd factor occurs before the final crossing, so its following state is greater than Y. Therefore $\prod_i (1 - u_i) \ge 1 - \sum_i u_i$ gives a lower product bound $1 - s/(2Y) \ge 1 - h/(2Y)$; the upper bound is one. □

For a tagged landing cell put

$$F_{M,Y}(h, y) = \#\{n \in I_M : \tau_Y(n) = h,\ T^h(n) = y\}. \tag{4.5}$$

Lemma 4.2 (odd-count rigidity)

If $h/(2Y) \le 1/3$, every source in a nonempty tagged fiber (h, y) has the same odd count.

Proof

If two sources had odd counts $s_1 < s_2$, put $A_i = 2^h y / 3^{s_i}$. Then $A_1 \ge 3A_2$, and (4.4) would give

$$\frac{n_1}{n_2} \ge 3\left(1 - \frac{h}{2Y}\right) \ge 2.$$

This is impossible for two integers in the half-open shell I_M, whose ratio is strictly smaller than two. □

Lemma 4.3 (tagged-fiber bound)

If $h/(2Y) \le 1/3$, then

$$F_{M,Y}(h, y) \le 1 + \frac{2h 2^M}{2Y - h}. \tag{4.6}$$

In particular, if $1 \le h \le H$ and $H/(2Y) \le 1/3$, then

$$F_{M,Y}(h, y) \le \frac{5}{2} H \frac{2^M}{Y}. \tag{4.7}$$

The estimates remain valid after arbitrary source restriction.

Proof

Lemma 4.2 fixes one odd count s. Put $A = 2^h y/3^s$ and $\varepsilon = h/(2Y)$. By (4.4), every source lies in $[(1-\varepsilon)A, A]$. Nonemptiness and $n < 2^{M+1}$ give

$$A < \frac{2^{M+1}}{1-\varepsilon}.$$

The interval contains at most $1 + \varepsilon A$ integers, proving (4.6). Under the second hypotheses, the nonconstant term is at most $(3/2)H\,2^M/Y$, while $1 \le H\,2^M/Y$, proving (4.7). □

Put $J_Y = (Y/2, Y] \cap \mathbb{N}$.

Proposition 4.4 (arbitrary-target linear transport)

Let $1 < Y < 2^M$, $H \ge 1$, and $H/(2Y) \le 1/3$. Every $B \subseteq J_Y$ satisfies

$$\#\{n \in I_M : \tau_Y(n) \le H,\ T^{\tau_Y(n)}(n) \in B\} \le \frac{5}{2} H^2 \frac{2^M}{Y} |B|. \tag{4.8}$$

The same inequality holds after arbitrary source restriction.

Proof

Every landing belongs to J_Y by (4.2). Since $Y < 2^M$, the passage time is positive. Sum (4.7) over the at most $H|B|$ tagged cells (h, y). □

5. A closed stopped map and its pullback

Fix

$$a_0 < r < 1, \qquad 0 < \eta < r - a_0, \tag{5.1}$$

and put $Y_M = 2^{\lfloor rM \rfloor}$. Choose M_0 so large that, for all $M \ge M_0$,

$$1 < Y_M < 2^M, \qquad \frac{M}{2Y_M} \le \frac{1}{3}, \qquad (a_0 + \eta)M + 1 + \eta \le \lfloor rM \rfloor. \tag{5.2}$$

If $n \in W_\eta \cap I_M$, then (3.6) gives

$$T^M(n) \le \varrho^M n^{1+\eta} < 2^{(a_0+\eta)M+1+\eta} \le Y_M. \tag{5.3}$$

Thus $\tau_{Y_M}(n) \le M$.

Put

$$N_0 = 2^{M_0}, \qquad \tilde{W}_\eta = W_\eta \cup [1, N_0). \tag{5.4}$$

Define a total map $G = G_{r,\eta} : \mathbb{N} \to \mathbb{N}$ and a block length $\ell : \mathbb{N} \to \mathbb{N}$ by

$$(G(n), \ell(n)) = \begin{cases} (T^{\tau_{Y_M}(n)}(n), \tau_{Y_M}(n)), & n \in W_\eta \cap I_M,\ M \ge M_0, \\ (n, 0), & \text{otherwise.} \end{cases} \tag{5.5}$$

Every $G(n)$ is the actual iterate $T^{\ell(n)}(n)$. On $\tilde{W}_\eta$,

$$0 \le \ell(n) \le \lfloor \log_2 n \rfloor, \qquad G(n) \le K_0 n^r, \qquad K_0 = N_0^{1-r}. \tag{5.6}$$

During every nontrivial block,

$$T^j(n) \le n^{1+\eta} \qquad (0 \le j \le \ell(n)). \tag{5.7}$$

For $S \subseteq \mathbb{N}$, define

$$\mathrm{FPPull}_{r,\eta}(S) = \tilde{W}_\eta \cap G^{-1}(S). \tag{5.8}$$

Theorem 5.1 (first-passage dense-set pullback)

Fix r, η as above and $0 < \chi < r$. There are constants $K_{FP}, D_c > 0$, depending only on these fixed parameters, such that, whenever $0 < D \le D_c$, the pullback (5.8) sends every (C, D)-dense set to a

$$(K_{FP}(C+1)D^{-2}, \chi D)\text{-dense set}. \tag{5.9}$$

Proof

By Proposition 3.3, $\tilde{W}_\eta$ is (C_η, D_η)-dense for fixed positive constants. Choose

$$D_c \le \min\{1, \frac{D_\eta}{\chi}\}, \qquad \chi D_c < 1. \tag{5.10}$$

Let S be (C, D)-dense, take $M \ge M_0$, and put $Y = Y_M$. A source in I_M outside (5.8) either lies outside W_η, or lies in W_η and lands in $B_M = S^c \cap J_Y$. Since

$$|B_M| \le \#(S^c \cap [1, Y]) \le C Y^{1-D}, \tag{5.11}$$

Proposition 4.4 with $H = M$ gives

$$\#\{n \in W_\eta \cap I_M : G(n) \notin S\} \le \frac{5}{2} C M^2 2^M Y^{-D}. \tag{5.12}$$

The floor costs only

$$Y^{-D} = 2^{-D\lfloor rM \rfloor} \le 2\, 2^{-rDM}. \tag{5.13}$$

Put $q = (r - \chi)\log 2 > 0$. For $0 < D \le 1$,

$$(M+1)^2 e^{-qDM} \le D^{-2}(DM+1)^2 e^{-qDM} \le C_q D^{-2}, \tag{5.14}$$

where $C_q = \sup_{x \ge 0}(x+1)^2 e^{-qx} < \infty$. Hence (5.12) is at most

$$K_1(C+1)D^{-2} 2^{(1-\chi D)M}. \tag{5.15}$$

The other bad sources satisfy

$$\#(W_\eta^c \cap I_M) \le C_\eta 2^{(M+1)(1-D_\eta)} \le K_2(C+1)D^{-2}2^{(1-\chi D)M} \tag{5.16}$$

by (5.10). Increase the fixed constant to absorb the finitely many shells below M_0. Summing the geometric shell series proves (5.9). The summation constant is uniform for $0 < D \le D_c$: if $a_D = 2^{1-\chi D}$, then

$$\sum_{M=0}^{J} a_D^M \le \frac{a_D}{a_D - 1} a_D^J, \qquad \frac{a_D}{a_D - 1} \le \frac{2}{2^{1-\chi D_c} - 1}.$$

Taking $2^J \le X < 2^{J+1}$ now gives the defining global bound (2.1) with exponent χD. □

Lemma 5.2 (height-sensitive first-passage clock)

Put

$$g = 1 - a_0 > 0.$$

For the parameters in (5.1), define

$$H_M = \left\lceil \frac{(1+\eta-r)M + 2 + \eta}{g} \right\rceil. \tag{5.17}$$

After increasing the fixed startup shell if necessary,

$$1 \le H_M \le M, \qquad \tau_{Y_M}(n) \le H_M \quad (n \in W_\eta \cap I_M), \tag{5.18}$$

and

$$H_M \le \frac{1+\eta-r}{g} M + \frac{2+\eta}{g} + 1. \tag{5.19}$$

The stopped map and the dense-set pullback theorem are unchanged when this shorter certified horizon replaces the coarse bound M.

Proof

The strict condition $\eta < r - a_0$ gives

$$\frac{1+\eta-r}{g} < 1,$$

so (5.17) is between (1) and (M) on all sufficiently large shells. The all-prefix envelope (3.6), the identity $\varrho^h = 2^{-gh}$, and $n < 2^{M+1}$ give

$$T^{H_M}(n) < 2^{-gH_M + (1+\eta)(M+1)} \le 2^{rM-1} \le 2^{\lfloor rM \rfloor} = Y_M. \tag{5.20}$$

The middle inequality is precisely the ceiling inequality in (5.17); the last uses $\lfloor rM \rfloor \ge rM - 1$. This proves (5.18), and the elementary upper bound for a ceiling proves (5.19).

Finally, Proposition 4.4 is already uniform in the horizon. Since $H_M \le M$, both

$$\frac{H_M}{2Y_M} \le \frac{1}{3} \quad \text{and} \quad \frac{5}{2} H_M^2 \frac{2^M}{Y_M} |B| \le \frac{5}{2} M^2 \frac{2^M}{Y_M} |B| \tag{5.21}$$

follow from the estimates used in Theorem 5.1. Thus no density exponent or pullback constant worsens. □

6. Bootstrap and proof of the main theorem

Fix

$$a_0 < r < 1, \qquad 0 < \eta < r - a_0, \qquad 0 < \chi < a_0. \tag{6.1}$$

Decrease the density exponent of $\tilde{W}_\eta$, if necessary, so that it is (C_0, D_0)-dense with $0 < D_0 \le D_c$. Define

$$S_0 = \tilde{W}_\eta, \qquad S_{j+1} = \mathrm{FPPull}_{r,\eta}(S_j). \tag{6.2}$$

Theorem 5.1 gives

$$D_{j+1} = \chi D_j, \qquad C_{j+1} \le K_{FP}(C_j + 1)D_j^{-2}. \tag{6.3}$$

Consequently

$$D_j = D_0\chi^j, \qquad \log(C_j + 2) = O(j^2). \tag{6.4}$$

For the second assertion, choose a fixed $A \ge \max\{1, K_{FP} + 2\}$. Since $D_j \le 1$,

$$C_{j+1} + 2 \le A(C_j + 2)D_j^{-2}.$$

Taking logarithms and summing $2\log(D_j^{-1}) = 2\log(D_0^{-1}) + 2j\log(1/\chi)$ proves (6.4).

If $n \in S_R$, put

$$n_0 = n, \qquad n_{i+1} = G(n_i) \quad (0 \le i < R). \tag{6.5}$$

Induction in (6.2) gives

$$n_i \in S_{R-i} \subseteq \tilde{W}_\eta \qquad (0 \le i \le R). \tag{6.6}$$

Thus (5.6) applies at every nonterminal stage, and a second induction gives

$$n_i \le K_0^{1+r+\cdots+r^{i-1}} n^{r^i} \le K_* n^{r^i}, \qquad K_* = K_0^{1/(1-r)}. \tag{6.7}$$

Every block is an actual Collatz segment, including the zero-step identity branch. Therefore

$$n_R = T^{k_R}(n), \qquad k_R = \sum_{i=0}^{R-1} \ell(n_i). \tag{6.8}$$

For the outer shell I_M, choose

$$R_M = \lceil \omega \log(M + 4) \rceil \tag{6.9}$$

with fixed $\omega > 0$, and define a varying-shell set H by

$$H \cap I_M = S_{R_M} \cap I_M. \tag{6.10}$$

Since $R_M \le \omega\log(M + 4) + 1$, (6.4) gives

$$D_{R_M} \ge D_0\chi(M + 4)^{-\omega\log(1/\chi)}, \qquad \log(C_{R_M} + 2) = O((\log M)^2). \tag{6.11}$$

The exceptional proportion in I_M is at most

$$2C_{R_M}\exp(-D_{R_M} M \log 2). \tag{6.12}$$

Put $\gamma = \omega\log(1/\chi)$. Equations (6.11)–(6.12) show that the logarithm of this proportion is at most

$$O((\log M)^2) - cM^{1-\gamma}. \tag{6.13}$$

It tends to $-\infty$ whenever

$$\omega\log(1/\chi) < 1. \tag{6.14}$$

After decreasing c and increasing the starting shell, the exceptional proportion is at most

$$2\exp(-c(M+4)^{1-\gamma}). \tag{6.15}$$

Lemma 2.1 proves that H has natural density one and supplies the global exceptional count.

On $H \cap I_M$, (6.7) and $R_M \geq \omega \log(M+4)$ give

$$\log n_{R_M} \leq \log K_* + (M+1)\log 2\,(M+4)^{-\omega\log(1/r)}. \tag{6.16}$$

Hence

$$n_{R_M} \leq \exp((\log n)^{1-\delta}) \tag{6.17}$$

for all sufficiently large M, provided

$$\delta < \omega\log(1/r). \tag{6.18}$$

Conditions (6.14) and (6.18) admit a common ω exactly when

$$\delta < \frac{\log(1/r)}{\log(1/\chi)}. \tag{6.19}$$

The time and intervening orbit are also controlled. From (5.6) and (6.7),

$$\ell(n_i) \leq \log_2 n_i \leq r^i \log_2 n + \log_2 K_*.$$

Summing the geometric progression and using $\log K_* = \log K_0/(1-r)$ gives

$$k_R \leq \frac{\log n}{(1-r)\log 2} + \frac{R\log K_0}{(1-r)\log 2}. \tag{6.20}$$

The same schedule admits a sharper raw Collatz clock. Let $s_h(x)$ be the number of odd shortcut states among $x, T(x), \dots, T^{h-1}(x)$. Direct induction gives the exact identity

$$\mathrm{Col}^{h+s_h(x)}(x) = T^h(x):$$

an even shortcut letter costs one raw step, while an odd shortcut letter is the two-step raw segment $x \mapsto 3x+1 \mapsto (3x+1)/2$. Odd counts are additive under concatenation. For a nonstartup retained block beginning at $n_i \in I_{M_i}$, the maximal barrier gives

$$s_{\ell(n_i)}(n_i) \leq \frac{\ell(n_i)}{2} + \frac{\eta M_i}{\log_2 3} \leq \frac{\ell(n_i)}{2} + \frac{\eta}{\log 3}\log n_i.$$

The startup identity branch contributes zero. Hence the raw time k_R^{raw} corresponding to (6.8) satisfies

$$\begin{aligned} k_R^{\mathrm{raw}} &\leq \frac{3}{2}\sum_{i<R}\ell(n_i) + \frac{\eta}{\log 3}\sum_{i<R}\log n_i \\ &\leq \frac{1}{1-r}\left(\frac{3}{2\log 2} + \frac{\eta}{\log 3}\right)\log n + O(R). \end{aligned} \tag{6.20r}$$

Here (6.7) gives $\sum_{i<R}\log n_i \leq (1-r)^{-1}\log n + O(R)$. For the schedule (6.9), the remainder in (6.20r) is $O(\log\log n) = o(\log n)$.

If $t_i = \sum_{m<i}\ell(n_m)$ and $t_i \leq j \leq t_{i+1}$, equations (5.7) and (6.7) give

$$T^j(n) \leq K_*^{1+\eta} n^{1+\eta}. \tag{6.21}$$

Proof of Theorem 1.1 and Corollaries 1.2–1.6

For $x = 1/7$, the positive-term series for $\log((1+x)/(1-x))$ gives

$$\log\frac{4}{3} > 2\left(\frac{1}{7} + \frac{1}{3\cdot 7^3}\right) = \frac{296}{1029}.$$

Consequently

$$\frac{1}{(1-a_0)\log 2} = \frac{2}{\log(4/3)} < \frac{1029}{148} < \frac{6953}{1000}. \tag{6.22}$$

The same rational lower bound also gives the strict raw-clock margin

$$\log\frac{4}{3} > \frac{296}{1029} > \frac{25}{87}, \qquad \frac{3}{\log(4/3)} < \frac{261}{25} = 10.44, \tag{6.22r}$$

because $296\cdot 87 - 25\cdot 1029 = 27 > 0$. Thus the strict interval

$$a_0 < r < 1 - \frac{1}{6.953\log 2} \tag{6.23}$$

is nonempty. As $r \downarrow a_0$ in this interval and $\chi \uparrow a_0$ through strict choices below a_0,

$$\frac{\log(1/r)}{\log(1/\chi)} \longrightarrow 1. \tag{6.24}$$

Given $0 < \delta < 1$, choose fixed r, χ so that (6.19) and (6.23) hold, and choose $0 < \eta < r - a_0$. Equations (6.8), (6.15), and (6.17) prove Theorem 1.1.

For Corollary 1.2, fix $0 < \sigma < 1 - \delta$. Since $\delta/(1-\sigma) < 1$, choose the same strict parameters with

$$q := \frac{\log(1/r)}{\log(1/\chi)}, \qquad \frac{\delta}{1-\sigma} < q.$$

Choose

$$\delta < \theta < q(1-\sigma), \qquad \omega = \frac{\theta}{\log(1/r)}.$$

Then the descent exponent is $\theta > \delta$, while the density-loss exponent is $\theta/q < 1-\sigma$. Thus (6.14), (6.18), and (6.15) hold with shell exponent σ, with the strict margin required by the stretched-exponential domination. Lemma 2.1 with $A = 2$ gives the prefactor 5 in (1.3).

For Corollary 1.3, (6.20), (6.23), and $R_M = O(\log M) = O(\log\log n)$ give

$$k_{R_M} < 6.953\log n$$

for every sufficiently large retained n. Given $\beta > 0$, choose $\eta < \min\{r - a_0, \beta\}$. The fixed factor in (6.21) is at most $n^{\beta-\eta}$ for all sufficiently large n, proving (1.6). For the raw clock, the leading coefficient in (6.20r) tends to $3/\log(4/3)$ as $r \downarrow a_0$ and $\eta \downarrow 0$. Equations (6.22r), (6.24), and the strict condition $\delta < 1$ therefore allow the same parameters to be chosen so that

$$\delta < \frac{\log(1/r)}{\log(1/\chi)}, \qquad \frac{1}{1-r}\left(\frac{3}{2\log 2} + \frac{\eta}{\log 3}\right) < \frac{261}{25}.$$

The strict margin absorbs the $o(\log n)$ term in (6.20r), proving the stated $10.44 \log n$ raw clock.

For Corollary 1.4, apply Corollary 1.3 with $\delta = 1/2$. If $\alpha > 0$, then $(\log n)^{-1/2} < \alpha$ for all sufficiently large n. Multiplication by the positive quantity $\log n$ gives

$$(\log n)^{1/2} < \alpha \log n.$$

After exponentiating,

$$\exp((\log n)^{1/2}) \leq \exp(\alpha \log n) = n^{\alpha}.$$

Thus the same witness supplied by Corollary 1.3 satisfies the fixed-power landing inequality without changing the $6.953 \log n$ clock.

For Corollary 1.5, fix $\alpha > 0$ and $0 < \sigma < 1$, and choose a fixed δ with

$$0 < \delta < 1 - \sigma.$$

Then

$$(\log n)^{1-\delta} = o(\log n),$$

so

$$\exp((\log n)^{1-\delta}) \leq n^{\alpha}$$

for every sufficiently large n. Thus, apart from finitely many initial values, the exceptional set in (1.7) is contained in the exceptional set in (1.3). Apply Corollary 1.2. Decreasing its positive constant and increasing the startup threshold absorb the finite initial contribution while retaining the prefactor 5: explicitly, replace its constant c_0 by $c_0/2$; then

$$5X\left(e^{-(c_0/2)(\log X)^{\sigma}} - e^{-c_0(\log X)^{\sigma}}\right) \longrightarrow \infty,$$

so it eventually dominates the fixed initial count. This proves (1.7).

For Corollary 1.6, put

$$c_* = \frac{1}{(1 - a_0)\log 2} = \frac{2}{\log(4/3)}.$$

Fix $0 < \alpha < 1$ and $\varepsilon > 0$. Choose $0 < \alpha' < \alpha$ with

$$c_*(\alpha - \alpha') < \frac{\varepsilon}{3}.$$

Choose a fixed integer $R \geq 1$ so large that

$$r = (\alpha')^{1/R} > a_0;$$

then $0 < r < 1$ and $r^R = \alpha'$. Finally choose $0 < \eta < r - a_0$ so small that

$$c_*(1 - \alpha')\frac{\eta}{1 - r} < \frac{\varepsilon}{3}. \tag{6.25}$$

Apply the fixed-depth version of (6.2) with this R. Repeated use of Theorem 5.1 shows that S_R is power-dense, hence has natural density one. Lemma 5.2 and (6.7) give, with a fixed horizon constant $C_h = (2+\eta)/(1-a_0)+1$,

$$\begin{aligned} k_R &\le \frac{1+\eta-r}{1-a_0}\sum_{i=0}^{R-1}\log_2 n_i + C_h R \\ &\le \frac{1+\eta-r}{(1-a_0)\log 2}\,\frac{1-r^R}{1-r}\log n + O_{\alpha,\varepsilon}(1) \\ &= c_*(1-\alpha')\left(1+\frac{\eta}{1-r}\right)\log n + O_{\alpha,\varepsilon}(1). \end{aligned} \tag{6.26}$$

The choices of α' and η leave at least $\varepsilon/3$ to absorb the fixed remainder, and hence give the clock in (1.8) for all sufficiently large n. At the terminal stage, (6.7) gives

$$n_R \le K_* n^{r^R} = K_* n^{\alpha'} \le n^{\alpha}$$

eventually, since $\alpha' < \alpha$. This proves Corollary 1.6. □

7. Scope

The argument proves an orbit-minimum statement on a set of natural density one. It does not prove descent for every starting value, exclude nontrivial cycles, or control the orbit after the selected witness. The endpoint $\delta = 1$ is a strict parameter limit and is not claimed. No finite computation is used as an all-depth premise.

The proof consumes only the parity-vector bijection, the maximal-barrier estimate, the first-passage reversal identity, and the resulting stopped-map pullback. In particular, no fixed-time endpoint-fiber moment or generated target equidistribution hypothesis is used.

Research and software disclosure

Generative-AI systems were used in proof exploration, adversarial auditing, finite diagnostics, formalization, and exposition. The author selected the theorem and proof architecture, reviewed the resulting artifacts, and accepts responsibility for the manuscript. The finite diagnostic is supporting evidence only and is not a premise of any theorem. The separate V2 Lean package formalizes the timed natural-density theorem, the quantitative exceptional counts in Corollaries 1.2 and 1.5, both clock refinements and the same-witness intermediate-orbit ceiling in Corollary 1.3, and the graded clock in Corollary 1.6. The manuscript proof remains self-contained and does not use the formal artifact as evidence. Its public `Main` module records the piecewise shortcut and raw Collatz maps, their literal iterates, the missing-count definition of natural density one, and the two descent predicates in a semantic dictionary; these are concrete definitions rather than abstract hypotheses of the formal theorems.

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